

Mapping Health: Analyzing Provider Distribution and Government Investment Across U.S. States, 2011–2013

CS 3891-05 / Project Final Presentation
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Objective

Examine relationships between healthcare provider density and government spending across U.S. states using network analysis

Goals

- Identify states that serve as central nodes or bridges in the healthcare network
- Evaluate correlations between provider distribution and state-level healthcare spending
- Study geographic patterns of healthcare resource distribution during ACA implementation period
- Identify key states where gaps in healthcare resources might exist

Description of Data

1. Office of the National Coordinator for HealthIT (ONC) (2011-2013)

- Healthcare provider distribution using data from ONC
- This dataset provider types were from all U.S. states, and aggregated across subdomains and primary care subdomains including:
 - Providers
 - Physicians
 - Nurse Practitioners
 - Physician Assistants
 - Primary care versions of these

2. State-level healthcare expenditure data from the Urban Institute compiled U.S. Census Bureau financial data:

- Total Hospital direct expenditure
- Per capita (USD) spent per person
- Adjusted in for inflation, real 2022\$

Provider counts are normalized and adjusted for population differences

- Providers per 100,000 residents for each state, using US Census Data

```
providers_per_100k = (total_provider_types / population) * 100000
```

- Both sets were normalized to account for state-level economic variations

```
providers_norm = (provider_values[idx] - provider_values.mean()) / provider_values.std()  
spending_norm = (spending_values[idx] - spending_values.mean()) / spending_values.std()
```

Description of Data

region	period	total provider types	population	providers per 100k	total hospital-dir expenditure
Alabama	2011	27480	4801627	572.31	353
Alabama	2012	28123	4817528	583.76	342
Alabama	2013	27098	4833722	560.6	376
• Rest of states below... • 2011-2013 periods were then averaged when doing edge calculations					

Methods - Geographic Network

Nodes: Each state is a node

- Geographically positioned using Albers projection coordinates
- Size indicates network centrality (combined measure of degree and betweenness centrality)
- Colors indicate census regions
 - Northeast: orange
 - Midwest: green
 - South: blue
 - West: pink

Edges: Healthcare system similarity connections between states

- Edge presence indicates strong healthcare pattern similarity (threshold: 0.7)
- Edge color indicates balance score strength (darker = stronger similarity)
- Edge thickness proportional to similarity strength (3x multiplier)

Balanced Score Calculation:

- Normalized provider density (providers per 100k population)
- Normalized hospital expenditure
- Enhanced contrast in balance score with exponential scaling
 - $\text{Score} = (1 / (1 + |\text{provider_diff} - \text{spending_diff}|))^1.5$
- Higher scores indicate more balanced healthcare approaches

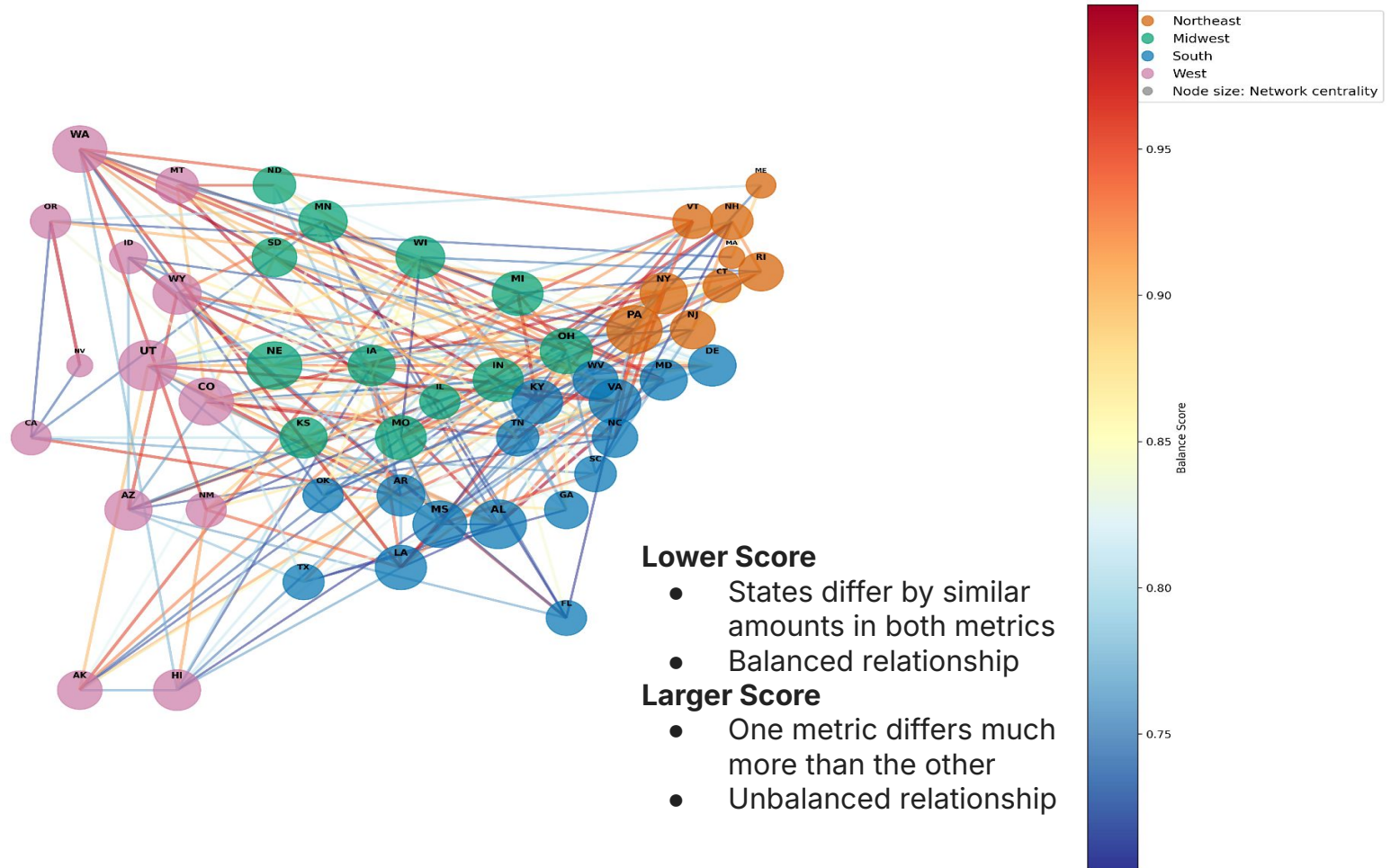
Network Properties:

- Undirected weighted graph
- Geographic layout preserving relative state positions
- Node sizes scaled 500-2500 based on importance
- Centrality weighted 70% degree, 30% betweenness

Visualization Features:

- Interactive colorbar showing balance score scale
- Regional grouping through node colors
- Size-adjusted state labels
- Edge opacity 0.6 for better visibility

Healthcare Network: Geographic Distribution of Provider-Spending Balance
Node size indicates network centrality; Darker edges show stronger balance



Methods - Combined Network

Nodes: Each state is a node

- Positioned using force-directed spring layout ($k=2.5$ spacing)
- Size indicates weighted centrality (40% degree, 30% betweenness, 30% eigenvector)
- Colors indicate census regions
 - Northeast: orange
 - Midwest: green
 - South: cyan
 - West: yellow

Edges: Healthcare system relationships with balance indication

- Edge presence shows strong similarity (threshold: 0.7)
- Edge color indicates provider-spending balance (red=provider dominant, blue=spending dominant)
- Edge thickness proportional to overall similarity strength (3x multiplier)
- Edge opacity set to 0.75 for readability

Balance Score Calculation:

- Normalized provider density and spending metrics
- Separate provider and spending similarity scores calculated
- Balance ratio (α) = $\text{provider_similarity} / \text{spending_similarity}$
- Combined similarity = $(\text{provider_sim} * \text{spending_sim} * \text{ratio_balance})^{(1/3)}$
- Log-scale transformation used for ratio visualization

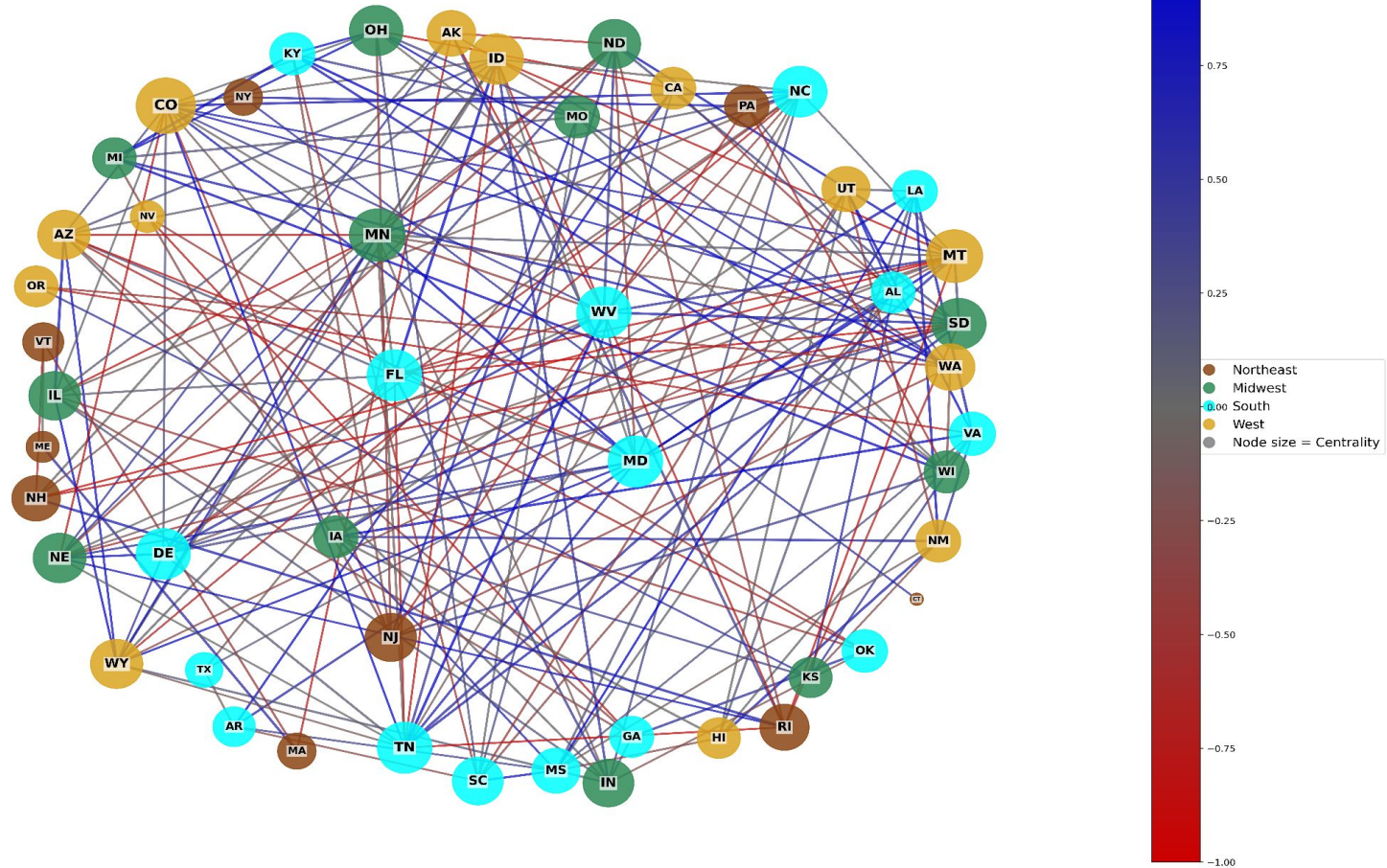
Network Properties:

- Undirected weighted graph
- Non-linear node size scaling for visual hierarchy
- Triple centrality measurement for comprehensive importance
- Simplified network used for centrality calculations

Visualization Features:

- Custom red-gray-blue colormap for balance indication
- White-bordered state labels for readability
- Balance ratio colorbar (-1 to +1 scale)
- Label sizes proportionally scaled with node sizes
- Detailed legend showing regions and centrality encoding

Healthcare Network: Combined Provider-Spending Similarity
Edge color indicates provider/spending balance
Edge thickness shows overall similarity strength



Methods - Clustered Network

Nodes: Each state is a node

- Node layout determined by spring force algorithm with weighted edges
- Size indicates combined centrality
(degree + betweenness + eigenvector) / 3
- Colors indicate census regions
 - Northeast: orange
 - Midwest: green
 - South: blue
 - West: pink

Edges: Two-tier healthcare similarity connections

- Weak connections: similarity score > 0.5
- Strong connections: similarity score > 0.8
- Edge thickness varies (1px weak, 3px strong)
- Edge opacity varies (0.3 weak, 0.8 strong)

Balanced Score Calculation:

- Higher scores indicate more balanced healthcare approaches
 - Same as Geographic balancing

Network Properties:

- Undirected weighted graph
- Spring layout with k=1.5 spacing
- Node sizes scaled 500-3500 based on importance
- Strong edges weighted 2x for layout calculation
- Centrality combines three measures equally
 - Degree centrality (local connections)
 - Betweenness centrality (bridge positions)
 - Eigenvector centrality (system-wide influence)

Visualization Features:

- Custom purple-white colormap for edges
- Community detection using Louvain algorithm
- Weak vs strong edge differentiation
- Regional color coding
- Detailed cluster statistics reporting:
 - States per cluster
 - Regional representation
 - Internal connection strength

Healthcare Network: Clustered Provider-Spending Balance
Darker edges indicate stronger balance, node size shows centrality



Network Comparison

Geographic

Combined

Clustered

Geographic: Prioritizes spatial accuracy

Combined: Prioritizes relationship visibility

Clustered: Prioritizes balance pattern

display

**Network
Layout**

Geographic: Single-tier, similarity-focused

Combined: Two-tier, relationship-focused

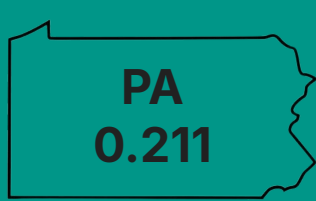
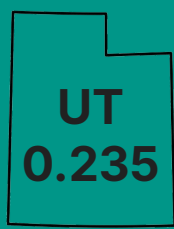
Clustered: Single-tier, balance-focused

**Edge
Structure**

**Centrality
Calculation**

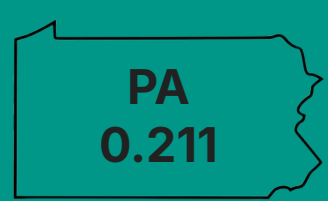
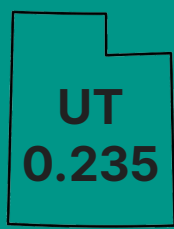
States Comparison

State	Population	Area	Capital	Language	Religion	Government	State Symbols
Alabama	5,024,278	70,542 sq mi	Montgomery	English	Christianity	Democratic	Yellowhammer, Yellowhammer, Yellowhammer
Alaska	731,449	663,300 sq mi	Juneau	English	Christianity	Democratic	Alaskan Malamute, Seal of the State of Alaska
Arizona	7,151,349	113,990 sq mi	Phoenix	English	Christianity	Democratic	Cactus, Saguaro, Saguaro
Arkansas	3,011,928	53,177 sq mi	Fayetteville	English	Christianity	Democratic	Apple, Apple, Apple
California	39,538,223	163,696 sq mi	Sacramento	English	Christianity	Democratic	Grizzly Bear, Grizzly Bear, Grizzly Bear
Colorado	7,011,219	104,140 sq mi	Denver	English	Christianity	Democratic	Buffalo, Buffalo, Buffalo
Connecticut	3,605,944	5,543 sq mi	Hartford	English	Christianity	Democratic	Grizzly Bear, Grizzly Bear, Grizzly Bear
Delaware	989,941	2,488 sq mi	Dover	English	Christianity	Democratic	Blue Hen, Blue Hen, Blue Hen
Florida	21,505,321	65,758 sq mi	Tallahassee	English	Christianity	Democratic	Orange, Orange, Orange
Georgia	10,787,332	59,725 sq mi	Atlanta	English	Christianity	Democratic	Wild Peach, Wild Peach, Wild Peach
Hawaii	1,415,872	10,931 sq mi	Honolulu	English	Christianity	Democratic	Pineapple, Pineapple, Pineapple
Idaho	1,840,642	83,542 sq mi	Boise	English	Christianity	Democratic	Shoshone, Shoshone, Shoshone
Illinois	12,812,232	57,914 sq mi	Springfield	English	Christianity	Democratic	State Seal, State Seal, State Seal
Indiana	6,785,528	36,420 sq mi	Indianapolis	English	Christianity	Democratic	Hoosier, Hoosier, Hoosier
Iowa	3,190,868	56,272 sq mi	Des Moines	English	Christianity	Democratic	State Seal, State Seal, State Seal
Kansas	1,781,039	81,813 sq mi	Topeka	English	Christianity	Democratic	State Seal, State Seal, State Seal
Kentucky	4,505,836	40,327 sq mi	Frankfort	English	Christianity	Democratic	State Seal, State Seal, State Seal
Louisiana	4,603,892	52,378 sq mi	Baton Rouge	English	Christianity	Democratic	State Seal, State Seal, State Seal
Maine	1,362,358	33,094 sq mi	Oxford	English	Christianity	Democratic	State Seal, State Seal, State Seal
Maryland	6,177,293	12,164 sq mi	Annapolis	English	Christianity	Democratic	State Seal, State Seal, State Seal
Massachusetts	7,001,421	10,354 sq mi	Springfield	English	Christianity	Democratic	State Seal, State Seal, State Seal
Michigan	10,076,331	96,716 sq mi	Lansing	English	Christianity	Democratic	State Seal, State Seal, State Seal
Minnesota	5,706,398	86,936 sq mi	St. Paul	English	Christianity	Democratic	State Seal, State Seal, State Seal
Mississippi	2,968,811	47,921 sq mi	Jackson	English	Christianity	Democratic	State Seal, State Seal, State Seal
Missouri	6,154,684	69,703 sq mi	Jefferson City	English	Christianity	Democratic	State Seal, State Seal, State Seal
Montana	1,080,577	147,040 sq mi	Helena	English	Christianity	Democratic	State Seal, State Seal, State Seal
Nebraska	1,961,590	77,348 sq mi	Lincoln	English	Christianity	Democratic	State Seal, State Seal, State Seal
Nevada	3,190,725	110,641 sq mi	Carson City	English	Christianity	Democratic	State Seal, State Seal, State Seal
New Hampshire	1,377,726	9,330 sq mi	Manchester	English	Christianity	Democratic	State Seal, State Seal, State Seal
New Jersey	9,288,594	19,277 sq mi	Trenton	English	Christianity	Democratic	State Seal, State Seal, State Seal
New Mexico	2,110,682	121,412 sq mi	Santa Fe	English	Christianity	Democratic	State Seal, State Seal, State Seal
New York	20,186,194	54,554 sq mi	Albany	English	Christianity	Democratic	State Seal, State Seal, State Seal
North Carolina	10,439,388	53,819 sq mi	Raleigh	English	Christianity	Democratic	State Seal, State Seal, State Seal
North Dakota	763,960	70,620 sq mi	Bismarck	English	Christianity	Democratic	State Seal, State Seal, State Seal
Ohio	11,796,449	44,826 sq mi	Columbus	English	Christianity	Democratic	State Seal, State Seal, State Seal
Oklahoma	3,959,351	69,562 sq mi	Norman	English	Christianity	Democratic	State Seal, State Seal, State Seal
Oregon	4,054,226	98,381 sq mi	Salem	English	Christianity	Democratic	State Seal, State Seal, State Seal
Pennsylvania	12,801,989	46,054 sq mi	Harrisburg	English	Christianity	Democratic	State Seal, State Seal, State Seal
Rhode Island	1,052,310	1,545 sq mi	Providence	English	Christianity	Democratic	State Seal, State Seal, State Seal
South Carolina	5,424,304	52,102 sq mi	Columbia	English	Christianity	Democratic	State Seal, State Seal, State Seal
South Dakota	886,667	77,116 sq mi	Sioux Falls	English	Christianity	Democratic	State Seal, State Seal, State Seal
Tennessee	6,918,721	62,082 sq mi	Nashville	English	Christianity	Democratic	State Seal, State Seal, State Seal
Texas	29,145,505	171,771 sq mi	Austin	English	Christianity	Democratic	State Seal, State Seal, State Seal
Utah	3,271,619	84,899 sq mi	Salt Lake City	English	Christianity	Democratic	State Seal, State Seal, State Seal
Vermont	643,045	9,616 sq mi	Winooski	English	Christianity	Democratic	State Seal, State Seal, State Seal
Virginia	8,631,383	42,775 sq mi	Richmond	English	Christianity	Democratic	State Seal, State Seal, State Seal
Washington	7,705,281	71,342 sq mi	Olympia	English	Christianity	Democratic	State Seal, State Seal, State Seal
West Virginia	1,853,470	62,058 sq mi	Charleston	English	Christianity	Democratic	State Seal, State Seal, State Seal
Wisconsin	5,893,714	65,498 sq mi	Madison	English	Christianity	Democratic	State Seal, State Seal, State Seal
Wyoming	576,853	97,813 sq mi	Cheyenne	English	Christianity	Democratic	State Seal, State Seal, State Seal



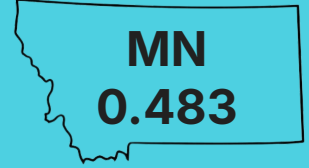
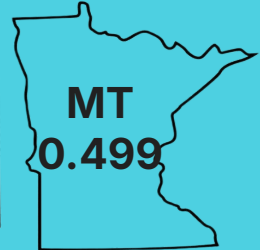
- Similar healthcare patterns to many of its neighboring states
- Consider physical proximity and healthcare similarities.

Geographic



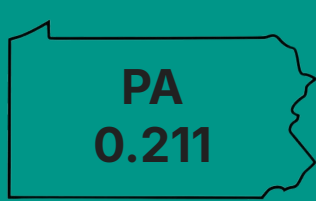
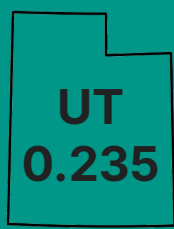
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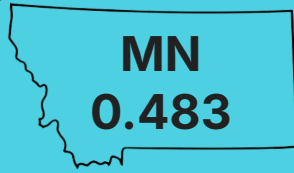
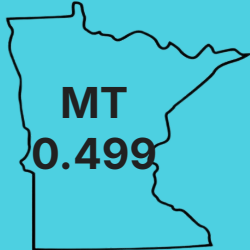
- States that balance provider and healthcare spending together successfully

Combined



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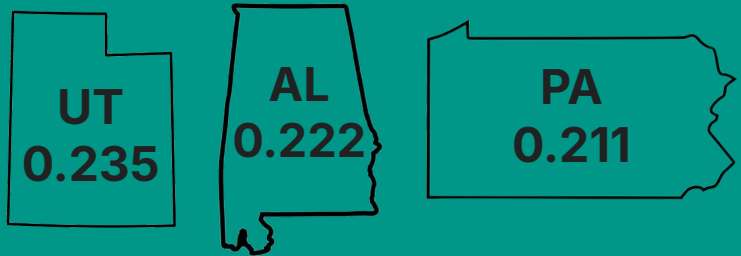
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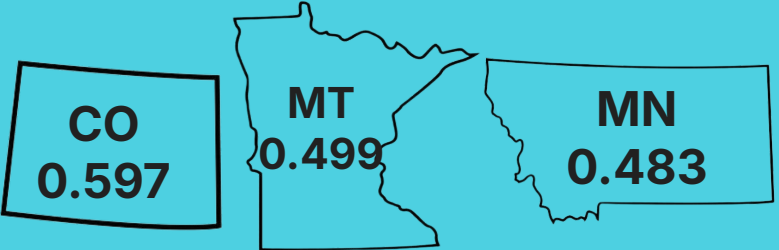
- Works well with many different approaches to healthcare
- Can share patterns between different groups of states that other states can learn from

Clustered



- Similar healthcare patterns to many of its neighboring states
- Consider physical proximity and healthcare similarities.

Geographic



- States that balance provider and healthcare spending together successfully

Combined



- Works well with many different approaches to healthcare
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Clustered



Top in Geo-Clust.

Bottom rankings across all networks

Midwest shows most consistent balance between the 3

Key States

Results - Summary Statistics

Measure	Providers (/100k)	Hospital-Dir Exp (2022\$)	Interpretation
Mean	687.76	228.32	Average values show typical provider density and spending levels. Provider density is more consistent.
Std Dev	92.42	172.16	Expenditure shows much higher variability relative to its mean compared to provider density.
Min	490.30	8.00	Extremely low minimum expenditure may indicate outliers or reporting differences.
Q1	620.33	74.75	Lower quartile shows baseline levels; expenditure has wider spread in lower values.
Median	681.27	166.00	Central values; expenditure median below mean indicates positive skew.
Q3	744.40	361.50	Upper quartile shows high-end typical values; large spread in expenditure.
Max	919.83	624.00	Maximum values show extremes; expenditure maximum is notably high relative to median.
Variance	8,541.24	29,637.35	Squared spread measure confirms much higher variability in expenditure.
Range	429.53	616.00	Total spread of values; expenditure shows wider relative range.
CV (%)	13.44	75.40	Relative variation measure shows expenditure is much more variable than provider density.
Skewness	0.43	0.50	Both slightly positively skewed, indicating some higher-than-typical values.
Kurtosis	-0.18	-1.07	Provider distribution near normal; expenditure shows flatter distribution with more extreme values.

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Results

1. Distribution Characteristics

- Provider distribution is nearly normal (kurtosis: -0.18).
- Expenditure distribution is notably flat (kurtosis: -1.07).
- Both distributions show slight positive skew (0.43 for providers, 0.50 for expenditure).

2. Regional Disparities

- Provider density varies moderately (coefficient of variation: 13.44%).
- Expenditure shows extreme variation (coefficient of variation: 75.40%).
- The range of expenditure (\$616) suggests significant regional differences.

3. Outliers and Extremes

- Minimum expenditure (\$8.00) requires further investigation.
- Maximum expenditure (\$624.00) is 3.8 times the median.
- Provider density shows a more consistent distribution compared to expenditure.

4. Policy Implications

- Provider distribution suggests effective regulatory standardization.
- Wide expenditure variation indicates differing regional approaches.
- There may be a need for spending standardization policies.

5. Statistical Relationships

- Different CV values suggest independent determining factors for provider density and expenditure.
- Expenditure shows more complex distribution patterns than provider density.
- Provider distribution is more predictable than spending patterns.

Conclusions

Findings

- Provider density consistency (CV = 13.44%) vs. spending variation (CV = 75.40%)
- Bridge states (Massachusetts and Minnesota) demonstrate successful balanced approaches
 - Found in multiple Top 5 Healthcare Systems lists
- Regional clustering exists, but cross-regional connections suggest opportunities for knowledge sharing
- Provider distribution follows near-normal pattern (kurtosis = -0.18)

Limitations

- 2011-2013 time frame does not reflect current healthcare landscape
- State-level analysis can mask intrastate/county variations
- No direct causation between spending patterns and healthcare outcomes
- Limited metrics/external factors misses important healthcare factors

Future Research

- Extend the time frame for better trends
- Incorporate geographic distance factors normalized for rural/urban populations
- Predictive modeling for healthcare outcomes across state clusters
- Analyze sub-state regional variations to capture local patterns

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Q&A